Communicating Mathematics in Writing: Quality Standards

Quality Standards for communicating mathematics in writing

Basic Quality Standards (applicable to all maths/AMS courses)

Sentences

This section taken from excerpts of Guide to Writing Mathematics, K.H. Law, University of Hong Kong.

1. Use complete sentences

One of the misconceptions students have about writing mathematics (which probably arises from their writing habits in secondary school) is that the writing should be composed mostly of equations and mathematical symbols. This is not true at all. By contrast, a piece of mathematical writing should contain mostly *words*, supplemented by equations and mathematical symbols. Take any mathematics textbook to verify this. Even in their worked examples, the solutions contain mostly words.

One of the very first basic principles in writing mathematics is therefore to write *structurally* in complete sentences. Take what you are reading now as an example. You can see that the whole subsection on 'using complete sentences' is divided into several *paragraphs*, each divided into several sentences. Each sentence is a *complete sentence*.

There can be equations and mathematical symbols, but still they form a complete sentence when properly read. Some examples are as follows.

Sentence containing equations and mathematical symbols	How it is read as a complete sentence
Hence $x+1>3$.	Hence x plus 1 is greater than 3.
Since $x>2$, it follows that $x^2+3x>2^2+3(2)=10$.	Since x is greater than 2, it follows that x squared plus three x is greater than two squared plus three times two which is equal to 10.
Thus $n\neq 0\ \forall n\in A$.	Thus n is not equal to 0 for all n that belongs to A.

2. Start sentences with capital letters

This sounds pretty much natural, as this is how you write an English passage. However, when it comes to writing mathematics, sometimes we may be tempted to start a sentence with a small letter because it is a symbol or a variable with which we want to use to start the sentence. Here are a couple of examples, and how they may be modified:

Sentence starting with a small letter	How it may be modified
p is not a prime number if it is divisible by 3 and greater than 3.	If the number p is divisible by 3 and greater than 3, then it is not a prime number.
$x^2+x+1=0$ has no real root since its discriminant is negative.	The equation $x^2+x+1=0$ has no real root since its discriminant is negative

Some even suggest that we should never start a sentence with a symbol or a variable, even if that is in capital letter. In that case similar modifications as above may be applied.

3. Do not overuse symbols

We conclude this section with a warning on using symbols. While using symbols correctly is essential for presenting mathematics accurately and concisely as we have shown throughout this section, there is always a danger of overusing them. For example, consider the following definition of a function $f(x)$ being continuous at the point a:

- $\forall \varepsilon > 0\ \exists \delta > 0$ s.t. $|x-a| < \delta \Rightarrow |f(x)-f(a)| < \varepsilon$.

While it is perfectly sound and correct, it is a bit difficult to read. Reducing the use of symbols would give

- For all $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x)-f(a)| < \varepsilon$ whenever $|x-a| < \delta$.

4. Avoiding isolated equations

We began this writing guide by saying that complete sentences should be used in mathematical writing. We conclude it by saying that complete *paragraphs* should be used. In short, this means we should avoid writing a few isolated equations without explaining the logical relationship between them. For example, consider the following:

$$5x + 1 = 16$$
$$5x = 15$$
$$x = 3$$

This is probably how primary and secondary school students 'write mathematics'. Indeed these can be considered as complete sentences. (Try to read them: '5 times x plus 1 is equal to 16. 5 times x is equal to 15. x is equal to 3.' Three complete sentences!) However when put together this makes no sense — what are we trying to say? Are we *assuming* the first equation holds? Or is it true that the first equation *does* hold because of some reason? Furthermore, what is the relationship between these equations? These are important issues and must be clarified by using proper connectives or symbols. For instance, two possible ways to connect these equations together are as follows:

1. According to the question, we have $5x + 1 = 16$. Since $5x + 1 = 16 \Leftrightarrow 5x = 15 \Leftrightarrow x = 3$, we conclude that the value of x is 3.

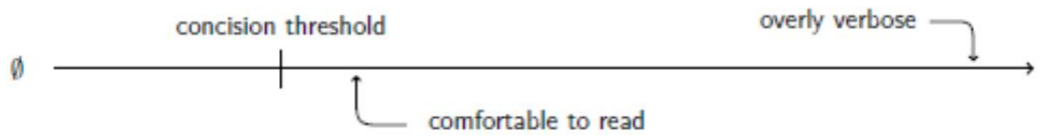
Concision

This and the following sections taken from University of Glasgow's SOLVE system

Mathematics is written concisely.

Steps of the proof or solution are clearly communicated with an appropriate level of verbosity. This means thinking about the intended audience (who you can normally reasonably assume is yourself before having completed the proof or solution) and unambiguously communicating enough detail to convince them without loosing the point in deatils. You do not need to include every step that you took to initially solve the problem, or any "dead-ends" you explored. You can reasonably expect that the reader will need to perform simple calculations to comprehend your writing. 1

There is a concision threshold that lies between writing nothing and an overly verbose submission where everything that is required to be stated is and nothing more (see Figure 1). Human mathematicians normally prefer to read immediately above this threshold, so that for example, signposting the type of proof (*by contradiction, by induction, etc.*) is included and mathematics written to teach concepts usually lies further above this concision threshold again. In automated theorem proving, computers usually work at the concision threshold. Without any "framing" of the arguement, it is challenging for to read and process mathematics presented in this way. Writing below the concision threshold will loose you marks for your mathematics independently of the communication mark.



Logic

Mathematics has a clear and logical structure.

Common errors include making statements like "If A $\implies$ B" when "A $\implies$ B" or "If A then B" is intended; the use of arrows to lead the eye around the page; equations with no linking text (hence, that is, thus, such that, etc.) stacked on the page one above the other.

As a sidenote, it is advisable not to use symbols that you are unsure how to "read out loud"! Reading mathematics invovles connecting the symbols with the underlying ideas

Variables

Symbols that identify mathematical objects are introduced and used appropriately.

For example, symbols used to identify variables are introduced before or as they are first used; symbols identifying vectors are underlined when handwritten and boldface when typeset; constants of integration are introduced as such.

Equality and comparison

Compatible mathematical objects are compared accurately and precisely.

Examples of comparing incompatible objects are, claiming a matrix is an element of $\mathbb{R}$, confusing a set with an element of a set, or using "$=$" where the appropriate word is "is".

Examples of inaccurate and/or imprecise comparisons include writing

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

when it is intended to claim they are row equivalent and '$\sim$' should have been used; or claiming something similar to

$$\sqrt{2} = 1.41421356237,$$

when we all know that $\sqrt{2}$ is irrational! Likely it was one of the following that was intended to be communicated:

$$\sqrt{2} = 1.41421356237 \text{ (11 d.p.), or } \sqrt{2} \approx 1.41421356237.$$

An illuminating examples is to ask yourself if the following statement is true or false:

$$\frac{1}{100} = \frac{1}{1000}.$$

Undoubtedly this question is ridiculous. However, it is equally ridiculous to state that an irrational number is rational.
Of course the above statement is false as written, nonetheless the statements

$$\frac{1}{100} = 0.0 \text{ (1 d.p.)}, \quad \frac{1}{1000} = 0.0 \text{ (1 d.p.)} \quad \text{and} \quad \frac{1}{100} \approx \frac{1}{1000}.$$

are true.